

Five-year vision: Molecule World Model

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Over the next five years, my plan is to build the *Molecule World Model*: a unified foundation model that learns joint representations of molecular sequences/structures/text and can reason/plan/act to accelerate scientific discovery. Our preliminary system, NatureLM¹ (46.7B parameters; trained on 143B tokens), shows strong scaling effect across 20+ scientific tasks. With NatureLM, one can chat with it to perform cross-domain tasks, just like “design a small molecule inhibitor for the protein $\langle \text{protein} \rangle$ MTKCGSTT... $\langle / \text{protein} \rangle$ ”, “Improve the BBB penetration ability and binding affinity of $\langle \text{mol} \rangle$ CN[C@@]1(CCCCC1=O)C2=CC=CC=C2Cl $\langle / \text{mol} \rangle$ on NMDA receptor”, “Design a material with bulk modulus 400 GPa”, *etc.* NatureLM has successfully designed compounds for tuberculosis (TB), malaria, and SARS-CoV-2, as well as DNA-binding proteins, which are all validated in wetlab experiments.

NatureLM demonstrates that a single, sequencebased science foundation model can unify small molecules, materials, proteins, and nucleic acids for crossdomain generation and optimization. Despite this progress, important gaps remain before we can deliver a reliable, endtoend scientific discovery engine. NatureLM can suggest what to make, but not why or how to test it, nor can they learn from the outcome.

To bridge this gap, we propose the *Molecule World Model*, a system that reconceptualizes scientific discovery as a closed-loop learning problem. Inspired by the “World Model” paradigm in general AI, our system will learn a generative model of the chemical and biological world, enabling it to simulate entire scientific strategies in silico before execution.

Advancing NatureLM to NatureLM*

The core of the molecule world model is an enhanced version of NatureLM, referred to as NatureLM*. NatureLM* is specifically designed to be physics- and structure-aware, enabling exceptional reasoning and cross-domain adaptability. Analogous to a vast scientific encyclopedia, its primary function is to understand and represent the static principles of the molecular world. It answers “what is” and “why is” questions by drawing on its deep, mechanistic model of material, chemistry and biology. This engine provides the universal context and foundational truth upon which all other system components are built. The architecture of NatureLM* is shown in Fig. 1(a).

Building NatureLM* requires addressing three critical technical challenges:

- Enhancing reasoning ability:** The primary goal is to improve verifiability, rationale quality, and alignment with expert preferences in the instructiontuning data. Specifically: (1) synthesize instruction–response pairs with automatic, toolgrounded verification (e.g., FEP, MD, DFT, Quantum Simulation; plus domain tools such as AlphaFold, MatterSim), and attach concise, explainable “why” rationales supported by citations and tool outputs; (2) collect expert preference data (pairwise or listwise) over rationales and multistep plans—not just final answers—to train preference and reward models aligned with scientific objectives; and (3) incorporate wetlab and autonomouslab supervision by establishing privacypreserving data feeds from partner labs, converting protocols and results into instruction–response exemplars with execution traces to enable closedloop learning under proper governance.
- Physical grounding:** Current research often emphasizes generating large-scale data through methods such as Density Functional Theory (DFT) or Molecular Dynamics (MD) and training neural networks to simulate these processes. We will utilize this data in two key ways: (1) to train encoders and decoders specific to the domain of the provided data (e.g., molecules, materials, proteins), as illustrated in Fig. 1; and (2) to train a reward model based on the provided data, using it as a feedback signal to regularize the training of NatureLM*.
- Cross-domain verification:** The focus is on tackling significant molecular science challenges that demand cross-domain knowledge. Key areas include, but are not limited to: Stable and Efficient Perovskite Solar Cell Design, High-Temperature Superconductor Discovery, Hard-to-Drug Cases, Molecular Glue Discovery, RNA Therapeutics, and Protein-Material Joint Design. Here the the principles for task selection: (1) Scientific Relevance: Tasks must address meaningful, real-world problems with the potential to advance molecular science, rather than serving as simple proof-of-concept experiments. (2) Foundation Readiness: An initial system or baseline capability should already be established, ensuring the task builds on existing progress rather than starting from scratch. (3) Interdisciplinary Impact: Tasks should leverage and integrate knowledge from multiple domains, demonstrating the ability of molecule world model to generalize across scientific fields.

¹<https://naturelm.github.io/>

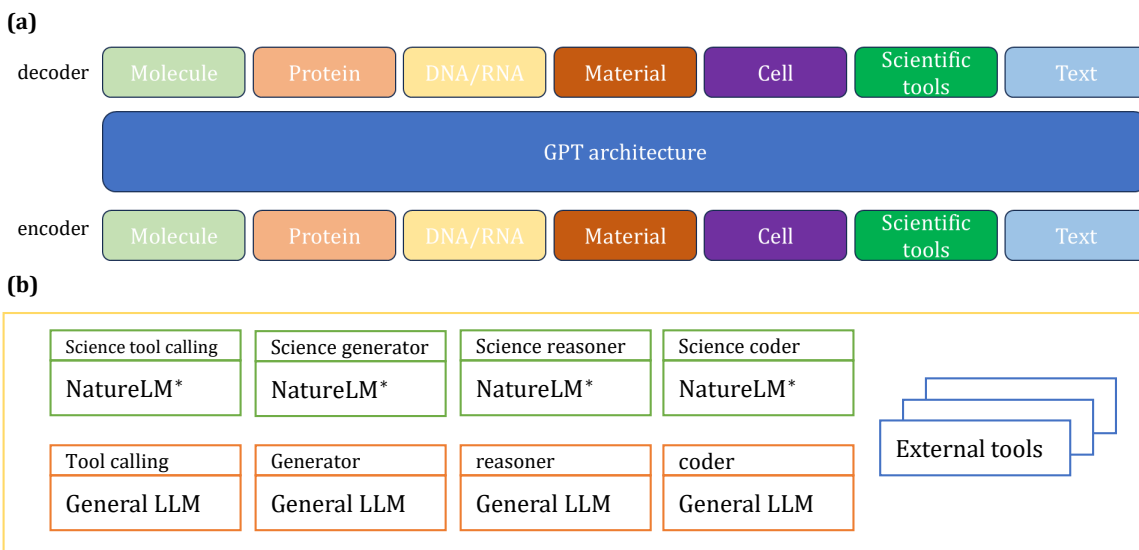


Figure 1: **Technical components of Molecule World Model.** (a) NatureLM* architecture. Extends coverage across domains compared to NatureLM. Structures of scientific entities can be encoded with VQVAE or diffusion models. Cellular states, typically represented as tabular measurements (e.g., genes and expression levels), are handled by dedicated modules. Tool-use trajectories (inputs, calls, outputs) are incorporated as a first-class modality to ground reasoning in scientific workflows. (b) Integration Framework: This diagram illustrates how NatureLM* integrates with a general-purpose Large Language Model (LLM) and external tools. This collaborative system enables the model to perform advanced functions and complex reasoning tasks, and could serve as the technical implementation of the dynamics engine and the planner.

The Molecule World Model: A Multi-Agent System for Scientific Discovery

While NatureLM* serves as a strong molecule foundation model, it is primarily a static repository of knowledge. To transition from understanding to action, and to create a true end-to-end scientific discovery engine, we propose the Molecule World Model. This model reconceptualizes discovery as a dynamic, closed-loop process, implemented as a sophisticated multi-agent system designed to generate actionable hypotheses and execute principled decision-making.

The Molecule World Model will be architected as a collaborative ecosystem of intelligent agents, where each agent has a specialized role, mirroring the structure of a human research team. The core agents include (Fig. 1(b)):

- Scientific Specialist Agents:** These agents are derived directly from NatureLM* through light fine-tuning or advanced prompting techniques. They are the domain experts of the system, responsible for tasks that require deep scientific knowledge. Their key capabilities include: (a) *Scientific Reasoning:* Analyzing complex problems in chemistry, biology, and materials science. (b) *Entity Generation:* Designing novel molecules, proteins, materials, and nucleic acids based on high-level goals. (c) *Scientific Code Generation:* Writing custom scripts to parse, filter, and analyze large datasets from computational experiments or public databases. This enables the agent to implement complex, ad-hoc filtration rules on a library of generated candidates. (d) *Tool Calling:* Interacting with specialized scientific software like DFT calculators, MD simulators, and docking programs.
- Generalist Agents:** Leveraging powerful general Large Language Models (LLMs), these agents act as project managers and communicators. They handle broader reasoning, planning, and user interaction. Their primary function is to decompose complex research questions into sub-tasks, delegate them to the appropriate specialist agents, and synthesize the results into a coherent response.
- Tool-Using Agents:** These are the system’s hands, providing a robust interface to the outside world. They execute commands to call external tools, from querying databases (e.g., Materials Project, PDB) and running computational simulations to, in the future, interfacing with automated wet-lab hardware.

The Core Workflow: Generating Actionable Hypotheses

The central function of the Molecule World Model is to move beyond simple generation and instead produce actionable hypotheses. This is achieved by unifying hypothesis generation and planning into a systematic workflow for proposing, testing, and executing scientific ideas. Each hypothesis generated by the system is a comprehensive, self-contained proposal designed for rigorous evaluation and includes four key elements:

1. *A Concise Claim*: A specific, testable statement that specifies the next concrete step to be taken (e.g., “Synthesizing a molecule *X* that can increase the binding affinity to target *Y* by inhibiting its allosteric site”, “Passivating the perovskite film with additive components to enhance solar cell stability by reducing charge recombination at defects.”).
2. *A Well-Founded Rationale*: A detailed justification for the claim, outlining the proposed scientific mechanisms or causal pathways. This rationale is grounded in the knowledge of NatureLM* and supported by references to prior work and data from tool outputs.
3. *Calibrated Confidence Levels*: An honest assessment of the hypothesis’s likelihood of success, explicitly highlighting potential uncertainties, risks, and failure modes. This allows researchers to make informed decisions about resource allocation.
4. *A Clear Execution Plan*: A step-by-step protocol for translating the hypothesis into a concrete experiment or computational task. This plan is detailed enough to be executed either by a human researcher or an automated lab.

Note that, our system is designed to generate a range of diverse hypotheses for a given problem, encouraging the exploration of multiple scientific avenues and reducing the risk of premature convergence on a single, potentially flawed idea.

Conclusions

Molecule world model will unite sequences, structures, and text with principled reasoning, physical constraints, and agentic execution. Our milestones, benchmarks, and governance plan ensure that advances are rigorous, transparent, and safe. Over five years, we will deliver validated case studies, standardized evaluations, and reusable tools that accelerate discovery in drugs, proteins, materials, etc. By fusing the creative power of generative AI with the predictive accuracy of physical simulation, the Molecule World Model will enable the systematic exploration of chemical space—from novel superconductors to next-generation perovskites, and from targeted cancer therapies to novel anti-inflammatory agents.

1. Yingce Xia, et al. (2025) "Nature Language Model: Deciphering the Language of Nature for Scientific Discovery", arXiv preprint